COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S3
DATE PERFORMED:	07 / 09 / 2026	**DATE SUBMITTED:**	09 / 14 / 2026
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ACTIVITY NO. 1.1 [Basic C++ Programming]

1. PROCEDURE OUTPUT

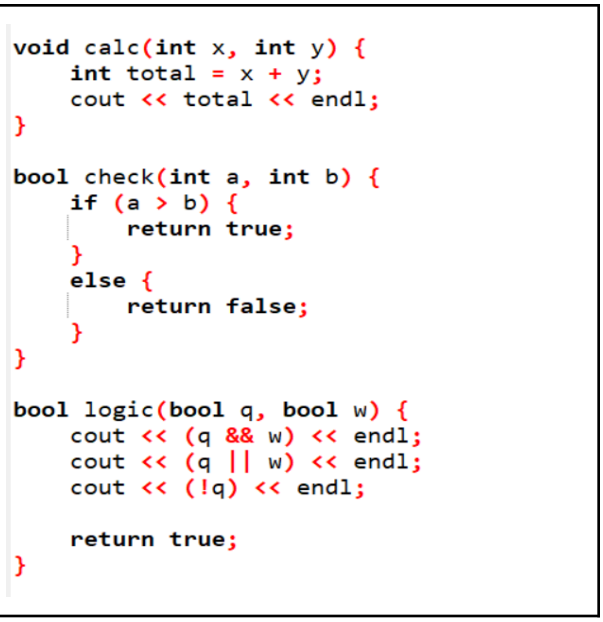
What procedures did you perform?

ILO A: Create Code That Follows the Basic C++ Code Structure

Sections	Answer
Header File Declaration Section	`#include <iostream>` `using namespace std;`

Global Declaration Section	`void calc(int x, int y);`
	`bool check(int a, int b);`
	`bool logic(bool q, bool w);`

Main Function	`int main() {`
	`calc(14, 36);`
	`bool ans = check(100, 250);`
	`cout << ans << endl;`
	`logic(true, false);`
	`return 0;`
	`}`

Method Definition	

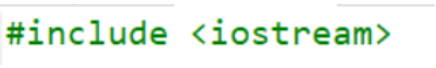
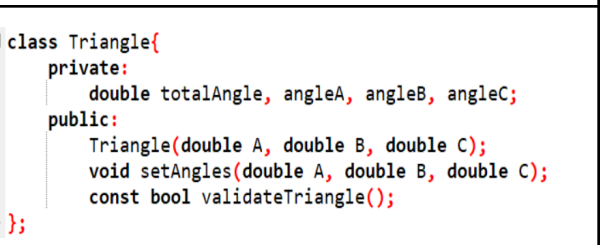
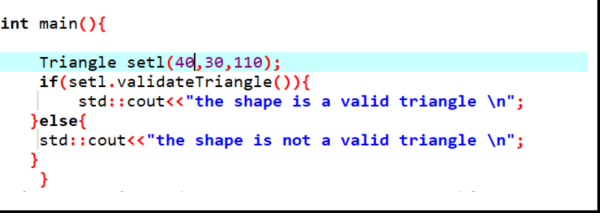
```
void calc(int x, int y) {
    int total = x + y;
    cout << total << endl;
}

bool check(int a, int b) {
    if (a > b) {
        return true;
    }
    else {
        return false;
    }
}

bool logic(bool q, bool w) {
    cout << (q && w) << endl;
    cout << (q || w) << endl;
    cout << (!q) << endl;

    return true;
}
```

ILO B:

Sections	Answer
Header File Declaration Section	
Global Declaration Section	
Class Declaration and Method Definition Section	
Main Function	

```
#include <iostream>
```

```
class Triangle{
    private:
        double totalAngle, angleA, angleB, angleC;
    public:
        Triangle(double A, double B, double C);
        void setAngles(double A, double B, double C);
        const bool validateTriangle();
};
```

```
int main(){

    Triangle set1(40,30,110);
    if(set1.validateTriangle()){
        std::cout<<"the shape is a valid triangle \n";
    }else{
    std::cout<<"the shape is not a valid triangle \n";
    }
     }
```

Method Definition	

```
Triangle::Triangle(double A, double B, double C){
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A+B+C;
}

void Triangle::setAngles(double A, double B, double C){
    angleA= A;
    angleB = B;
    totalAngle = A+B+C;
}
const bool Triangle::validateTriangle(){
    return(totalAngle <= 180);
}
```

Table 1 - 1. C++ Structure code for Answer

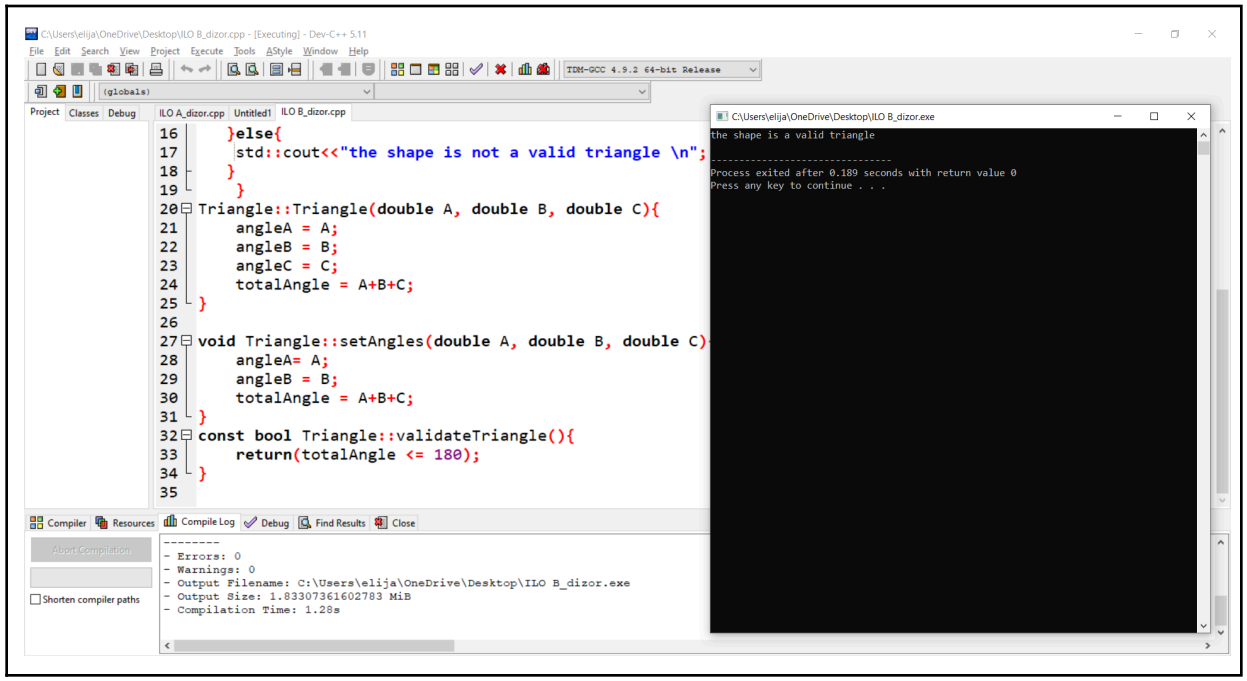


Table 1 - 2. ILO B output observations and comments.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

ILO C: Solve Different Problems using the C++ Programming Language.

1. Create a C++ program to swap the two numbers in different variables.

Code	Output

```cpp
#include <iostream>
using namespace std;

int main() {
    int num1 = 15;
    int num2 = 30;

    int temp = num1;
    num1 = num2;
    num2 = temp;

    cout << "" << num1 << endl;
    cout << "" << num2 << endl;

    return 0;
}
```

```
C:\Users\elija\OneDrive\Desktop\ILOC problem 1.exe
30
15

--------------------------------
Process exited after 0.1833 seconds with return value 0
Press any key to continue . . .
```

2. Create a C++ program that has a function to convert temperature in Kelvin to Fahrenheit.

Code	Output

```cpp
#include <iostream>
using namespace std;

double convertTemp(double kelvin);

int main() {
    double k = 300.0;
    double f = convertTemp(k);

    cout << "Fahrenheit is " << f << endl;
    return 0;
}

double convertTemp(double kelvin) {

    double fahrenheit = (kelvin - 273.15) * 1.8 + 32.0;
    return fahrenheit;
}
```

```
Select C:\Users\elija\OneDrive\Desktop\ILOC problem 2.exe
ahrenheit is 80.33

------------------------------
rocess exited after 0.2578 seconds with return value 0
ress any key to continue . . .
```

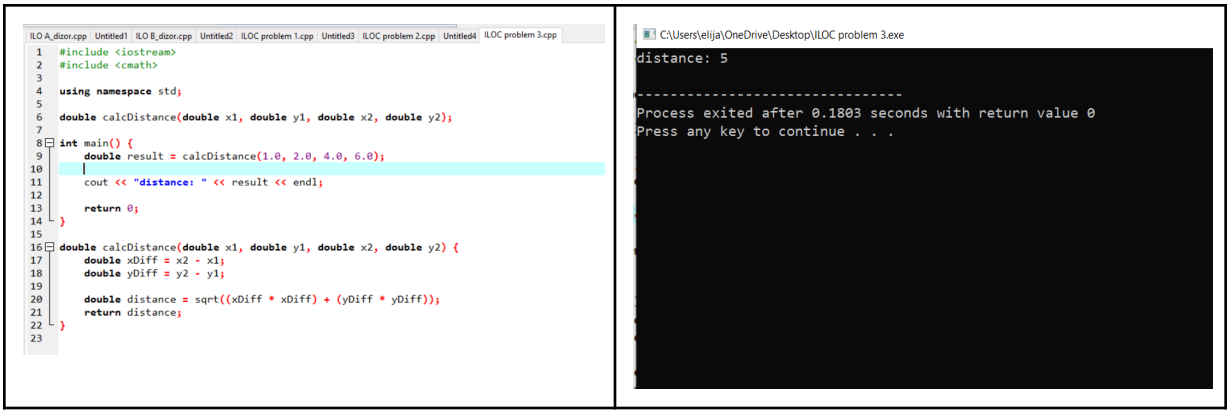
3. Create a C++ program that has a function that will calculate the distance between two points.

4. Modify the code given in ILO B and add the following functions:
a. A function to compute the area of a triangle

b. A function to compute for the perimeter of a triangle

c. A function that determines whether the triangle is acute-angled, obtuse-angled or 'others.'

Code	Output



```cpp
#include <iostream>
#include <cmath>

using namespace std;

double calcDistance(double x1, double y1, double x2, double y2);

int main() {
    double result = calcDistance(1.0, 2.0, 4.0, 6.0);

    cout << "distance: " << result << endl;

    return 0;
}

double calcDistance(double x1, double y1, double x2, double y2) {
    double xDiff = x2 - x1;
    double yDiff = y2 - y1;

    double distance = sqrt((xDiff * xDiff) + (yDiff * yDiff));
    return distance;
}
```

```
C:\Users\elija\OneDrive\Desktop\ILOC problem 3.exe
distance: 5

--------------------------------
Process exited after 0.1803 seconds with return value 0
Press any key to continue . . .
```

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **chatgpt**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Code assistance, explanation of concepts**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I used ai to help me understand the code difference between python and C++ and what are their similarities and functions**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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"I affirm that I will not give or receive any unauthorized help on this activity/exam and that all work will be my own."